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Dilpreet Singh Brar, *[a] Birmohan Singh[b] and Vikas Nanda[a]

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Ensuring food safety and sustainability is essential in the global spice industry. This study presents an AI-driven framework to detect red chili powder (RcP) adulteration, enhancing food quality assessment using DenseNet-121 and DenseNet-169 with the AdamClr optimizer. The model accurately identifies contaminants like wheat bran, rice hulls, and wood sawdust, improving consumer safety and regulatory compliance. This approach promotes ethical food production, minimizes economic losses, and reduces the environmental impact of food fraud. explainable AI (XAI) through Grad-CAM ensures transparency, fostering stakeholder trust. Additionally, the model's efficiency optimizes testing and quality control, supporting sustainable supply chains. By detecting adulteration and ensuring RcP purity, this study advances sustainable food security. The AI-powered method revolutionizes quality assurance in the spice industry and establishes a foundation for future AI applications in food fraud detection, reinforcing global efforts for a safer and more sustainable food system.

1. Introduction

Artificial Intelligence (AI) is revolutionising the food sector from farm to fork by enhancing production methods, streamlining supply chain management, advancing quality assurance protocols, and strengthening consumer safety.[1] Moreover, as the food industry addresses critical challenges such as food safety, waste reduction, labour shortages, and evolving consumer demands, the AI-driven food technology market is projected to reach USD 27.73 billion by 2029.[2] This growth is propelled by the demand for tailored nutrition, stricter safety standards, supply chain optimisation, and sustainability. These changes create a new era for producing, distributing, and consuming food worldwide.[3] Additionally, there remains a long way to go to fully advance the application of AI technology in food quality evaluation, particularly in the accurate detection of adulteration. This progress can be accelerated by focusing on specific sectors, such as detecting adulteration in spices and refining AI-based technology to address the challenges.[4] By systematically

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An XAI-enabled 2D-CNN model for non-destructive detection of natural adulterants in the wonder hot variety of red chilli powder†

Dilpreet Singh Brar, *[a] Birmohan Singh[b] and Vikas Nanda[a]

AI revolutionizes the food sector by improving production, supply chains, quality assurance, and consumer safety. Therefore, this work addresses the alarming issue of red chilli powder (RcP) adulteration, with the introduction of an AI-driven framework for RcP adulteration detection, leveraging an empirical evaluation of DenseNet-121 and 169. To optimize convergence and enhance the performance, the AdamClr optimizer was incorporated, in a learning rate range between 0.00005 and 0.01. Two datasets (DS I and DS II) were developed for evaluation of DenseNet models. DS I consists of two classes: Class 1 (Label = C1_PWH) representing pure RcP (variety = Wonder Hot (WH)) and Class 2 (Label = C2_AWH) containing samples adulterated with five natural adulterants (wheat bran (WB), rice hull (RB), wood saw (WS), and two low-grade RcP), whereas DS II comprises 16 classes, including one class of pure RcP and 15 classes representing adulterated RcP with varying concentrations of the five adulterants (each at 5%, 10%, and 15% concentration). For binary classification (for DS I), DenseNet-169 at batch size (BS) 16 delivered an accuracy of 99.99%, while, in multiclass classification (for DS II) for determination of the percentage of adulterant, DenseNet-169 at BS 64 produced the highest accuracy of 95.16%. Furthermore, Grad-CAM explains the DenseNet-169 predictions, amd the obtained heatmaps highlighting the critical regions influencing classification decisions. The proposed framework demonstrated high efficacy in detecting RcP adulteration in binary as well as multiclass classification. Overall, DenseNet-169 and XAI present a transformative approach for enhancing quality control and assurance in the spice industry.

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